

## **Annex D**

All the signals related to the 10 MW BESS as per the Typical Signal List “Annex D1- Typical Signal List for BESS” shall be configured in a Gateway as per the IEC 60870 – 5 – 104 protocol and made available to NSCC before the commissioning by the BESS developer via the CEB communication equipment at adjacent CEB Grid Substation through a secured communication link using fiber optic cable.

The communication interfaces shall be configured as in Annex D2. Finalized Signal List shall be submitted at least 02 weeks prior to the commissioning.

Configuration of NSCC SCADA Master Station for integration of the signals shall be done by the CEB. The integrated signals shall be tested up to the NSCC by the BESS developer with coordinating NSCC for relevant witnessing.

BESS can be managed by the NSCC as per the requested signals in Annex D1. Therefore, the separate remote-control level for NSCC operations and interlocking operations shall be activated for the NSCC operations at the BESS control center. Real time operations shall be arranged for commissioning and simulations of any signals shall not be accepted.

The Single Line diagram of the BESS attaching to the CEB GS shall be submitted at the design stage to finalize the NSCC SCADA requirement.”

| Signal Description                                           | Type Of Signal | Signal Type Description |                                 |                 | Description          | BESS |
|--------------------------------------------------------------|----------------|-------------------------|---------------------------------|-----------------|----------------------|------|
|                                                              |                | 104 TYPE ID             | IEC 60870 - 5 - 104 DESCRIPTION | USER DATA CLASS |                      |      |
|                                                              |                |                         |                                 |                 | <b>ACTION</b>        |      |
| <b>Status Indications</b>                                    |                |                         |                                 |                 |                      |      |
| Circuit Breaker Open/Close                                   | DPI            | 31                      | M_DP_TB_1                       | CLASS 1         | Open/Close           | 2    |
| Bus Isolator Open/Close                                      | DPI            | 31                      | M_DP_TB_1                       | CLASS 2         | Open/Close           | 2    |
| Line Isolator Open/Close                                     | DPI            | 31                      | M_DP_TB_1                       | CLASS 2         | Open/Close           | 2    |
| Earth Switch Open/Close                                      | DPI            | 31                      | M_DP_TB_1                       | CLASS 2         | Open/Close           | 2    |
| Plant Control NSCC/Plant                                     | DPI            | 31                      | M_DP_TB_1                       | CLASS 2         | NSCC/Plant           | 2    |
| Operating Mode 1                                             | SPI            | 30                      | M_SP_TB_1                       | CLASS 1         | Enable/Disable       | 1    |
| Operating Mode 2                                             | SPI            | 30                      | M_SP_TB_1                       | CLASS 1         | Enable/Disable       | 1    |
| Operating Mode n                                             | SPI            | 30                      | M_SP_TB_1                       | CLASS 1         | Enable/Disable       | 1    |
| Active Power Controller Mode                                 | SPI            | 30                      | M_SP_TB_1                       | CLASS 1         | Enable/Disable       | 1    |
| Reactive Power Controller Mode                               | SPI            | 30                      | M_SP_TB_1                       | CLASS 1         | Enable/Disable       | 1    |
| Voltage Controller Mode                                      | SPI            | 30                      | M_SP_TB_1                       | CLASS 1         | Enable/Disable       | 1    |
| Power Factor Controller Mode                                 | SPI            | 30                      | M_SP_TB_1                       | CLASS 1         | Enable/Disable       | 1    |
| Battery Charging/Discharging status                          | SPI            | 30                      | M_SP_TB_1                       | CLASS 1         | Charging/Discharging | 1    |
| Communication Failure Alarm                                  | SPI            | 30                      | M_SP_TB_1                       | CLASS 1         | Alarm/Normal         | 1    |
| Plant Available / Unavailable                                | SPI            | 30                      | M_SP_TB_1                       | CLASS 1         | Alarm/Normal         | 1    |
|                                                              |                |                         |                                 |                 |                      |      |
| <b>Measurements</b>                                          |                |                         |                                 |                 |                      |      |
| Active Power (MW)                                            | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | MW                   | 1    |
| Reactive Power (MVar)                                        | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | MVar                 | 1    |
| Current (A)                                                  | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | A                    | 1    |
| Voltage (kV)                                                 | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | kV                   | 1    |
| Power Factor                                                 | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | PF                   | 1    |
| Frequency (Hz)                                               | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | Hz                   | 1    |
| Today Total Charging Active Energy                           | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | MWh                  | 1    |
| Today Total Discharging Active Energy                        | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | MWh                  | 1    |
| Yesterday Total Charging Active Energy                       | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | MWh                  | 1    |
| Yesterday Total Discharging Active Energy                    | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | MWh                  | 1    |
| Today total Lagging Reactive Energy                          | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | MVarh                | 1    |
| Today total Leading Reactive Energy                          | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | MVarh                | 1    |
| Yesterday total Lagging Reactive Energy                      | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | MVarh                | 1    |
| Yesterday total Leading Reactive Energy                      | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | MVarh                | 1    |
| Plant MW Availability - Charging                             | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | MW                   | 1    |
| Plant MW Availability -Discharging                           | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | MW                   | 1    |
| Battery State of Charge (SoC)                                | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | %                    | 1    |
| Maximum Available Discharging Energy                         | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | MWh                  | 1    |
| Maximum Available Charging Energy                            | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | MWh                  | 1    |
| Ambient Temperature                                          | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | °C                   | 1    |
| Charge Ramp rates                                            | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | MW/min               | 1    |
| Discharge Ramp rates                                         | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | MW/min               | 1    |
| Number of Used round trip cycles per year at Boundary        | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | Count                | 1    |
| Number of Remaining round trip cycles per year at Boundary   | AI             | 13                      | M_ME_NC_1                       | CLASS 2         | Count                | 1    |
|                                                              |                |                         |                                 |                 |                      |      |
| <b>Control Commands</b>                                      |                |                         |                                 |                 |                      |      |
| Select Operating Mode 1                                      | SPC            | 45                      | C_SC_NA_1                       | CLASS 1         | Activate/Deactivate  | 1    |
| Select Operating Mode 2                                      | SPC            | 45                      | C_SC_NA_1                       | CLASS 1         | Activate/Deactivate  | 1    |
| Select Operating Mode n                                      | SPC            | 45                      | C_SC_NA_1                       | CLASS 1         | Activate/Deactivate  | 1    |
| Select Active Power Control                                  | SPC            | 45                      | C_SC_NA_1                       | CLASS 1         | Activate/Deactivate  | 1    |
| Select Reactive Power Control                                | SPC            | 45                      | C_SC_NA_1                       | CLASS 1         | Activate/Deactivate  | 1    |
| Select Voltage Control                                       | SPC            | 45                      | C_SC_NA_1                       | CLASS 1         | Activate/Deactivate  | 1    |
| Select Power Factor Control                                  | SPC            | 45                      | C_SC_NA_1                       | CLASS 1         | Activate/Deactivate  | 1    |
|                                                              |                |                         |                                 |                 |                      |      |
| Active Power Set Point Command ( + Discharging / - Charging) | AO             | 50                      | C_SE_NC_1                       | CLASS 1         | MW                   | 1    |
| Reactive Power Set point Command ( + Lagging / - Leading)    | AO             | 50                      | C_SE_NC_1                       | CLASS 1         | MVar                 | 1    |
| Voltage Set Point Command                                    | AO             | 50                      | C_SE_NC_1                       | CLASS 1         | kV                   | 1    |
| Power Factor Set Point Command                               | AO             | 50                      | C_SE_NC_1                       | CLASS 1         | PF                   | 1    |
| Charge Ramp rates                                            | AO             | 50                      | C_SE_NC_1                       | CLASS 1         | MW/min               | 1    |
| Discharge Ramp rates                                         | AO             | 50                      | C_SE_NC_1                       | CLASS 1         | MW/min               | 1    |
|                                                              |                |                         |                                 |                 |                      |      |

**Legends:**

DPI - Double Point Indication

SPI - Single Point Indication

DPC - Double Point Control Command

SPC - Single Point Control Command

**Communication Path Configurations for the Gateway installing at BESS Control Center under the Developer Premises**

**a) SCADA Protocol -IEC 60870-5-104**

- Speed of the TCP/IP connection : 10/100Mbps
- Gateway IP address : will be provided at commissioning period.
  - IP address of the Master Station at NSCC : 10.3.1.8 & 10.3.1.9
  - IP address of the Master Station at BNSCC : will be provided at commissioning period.
  - Subnet Mask : 255.255.255.240

➤ **Application Layer**

Fixed lengths are defined in the protocol as below.

- Common Address of ASDU : 2 octets
- Information object address : 3 octets
- Cause of transmission : 2 octets
- Length of APDU : Maximum length of APDU per system (253)